

Jeremy G. Baier

Department of Physics, Oregon State University

305 Weniger Hall Corvallis, OR 97331-6507, USA

☎ +1 (504) 913 4014 • ✉ baierj@oregonstate.edu • 🌐 [jeremy-baier](https://www.jeremy-baier.com)
 🐦 [jeremygbaier](https://twitter.com/jeremygbaier) • in [jeremygbaier](https://www.linkedin.com/in/jeremygbaier)

Publications

Published.....

29. *The NANOGrav 15 yr and 20 yr Datasets: Timing Events and Pulse Shape Changes.*
 Ben Jacobson-Bell, [...], **J. G. Baier**, et al. [64 Authors]
[The Astrophysical Journal](#), **1005**, 1, (2026)
28. *The NANOGrav 15 yr Dataset: Customized Chromatic Noise Models.*
 Bjorn Larsen, **J. G. Baier**, et al. [56 Authors]
[The Astrophysical Journal](#), **1005**, 1, (2026)
27. *The NANOGrav 15 yr Data Set: Piecewise Power-law Reconstruction of the Gravitational-wave Background.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [111 Authors]
[The Astrophysical Journal Letters](#), **1004**, 2, (2026)
26. *The NANOGrav 12.5 yr Dataset: Chromatic Noise Characterization and Mitigation with Time-domain Kernels.*
 Jeffrey S. Hazboun, [...], **J. G. Baier**, et al. [67 Authors]
[The Astrophysical Journal](#), **1003**, 1, (2026)
25. *Deep Search for Joint Sources of Gravitational Waves and High-energy Neutrinos with IceCube during the Third Observing Run of LIGO and Virgo.*
 R. Abbasi, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **1003**, 1, (2026)
24. *Inference on inner galaxy structure via gravitational waves from supermassive binaries.*
 Yifan Chen, [...], **J. G. Baier**, et al. [111 Authors]
[Nature Astronomy](#), **10**, pp. 554-563, (2026)
23. *Gravitational Wave Measurement of the $M_{\text{BH}}-M_{\text{bulge}}$ Intrinsic Scatter at High Redshift.*
 Cayenne Matt, [...], **J. G. Baier**, et al. [112 Authors]
[Arxiv:2603.11167](#)
22. *The NANOGrav 15 yr Dataset: Targeted Searches for Supermassive Black Hole Binaries.*
 Nikita Agarwal, [...], **J. G. Baier**, et al. [122 Authors]
[The Astrophysical Journal Letters](#), **998**, 1, (2026)
21. *Inferring $M_{\text{BH}}-M_{\text{bulge}}$ Evolution from the Gravitational-wave Background.*
 Cayenne Matt, [...], **J. G. Baier**, et al. [107 Authors]
[The Astrophysical Journal](#), **997**, 2, (2026)
20. *Rapid construction of joint pulsar timing array data sets: the Lite method.*
 B. Larsen, [...], **J. G. Baier**, et al. [43 Authors]
[Monthly Notices of the Royal Astronomical Society](#), **542**, 4, (2025)
19. *The NANOGrav 15 yr Data Set: Search for Gravitational-wave Memory.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [105 Authors]
[The Astrophysical Journal](#), **987**, 1, (2025)
18. *Search for Gravitational Waves Emitted from SN 2023ixf.*
 A. G. Abac, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **985**, 2, (2025)

17. *The NANOGrav 15 yr Data Set: Harmonic Analysis of the Pulsar Angular Correlations.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [107 Authors]
[The Astrophysical Journal](#), **985**, 1, (2025)
16. *A sensitivity curve approach to tuning a pulsar timing array in the detection era.*
J. G. Baier, Jeffrey S. Hazboun, Joseph D. Romano
[Classical and Quantum Gravity](#), **42**, 7, (2025)
15. *Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run.*
 A. G. Abac, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **983**, 2, (2025)
14. *Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run.*
 Gayathri Raman, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **980**, 2, (2025)
13. *The NANOGrav 15 yr dataset: Posterior predictive checks for gravitational-wave detection with pulsar timing arrays.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [104 Authors]
[Physical Review D](#), **111**, 4, (2025)
12. *The NANOGrav 15 yr Data Set: Looking for Signs of Discreteness in the Gravitational-wave Background.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [100 Authors]
[The Astrophysical Journal](#), **978**, 1, (2025)
11. *The NANOGrav 15 Yr Data Set: Removing Pulsars One by One from the Pulsar Timing Array.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [105 Authors]
[The Astrophysical Journal](#), **978**, 2, (2025)
10. *The NANOGrav 15 yr Data Set: Running of the Spectral Index.*
 Gabriella Agazie, [...], **J. G. Baier**, et al. [105 Authors]
[The Astrophysical Journal Letters](#), **978**, 2, (2025)
9. *A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154.*
 A. G. Abac, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **977**, 2, (2024)
8. *Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo.*
 A. G. Abac, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **973**, 2, (2024)
7. *Ultralight vector dark matter search using data from the KAGRA O3GK run.*
 A. G. Abac, [...], **J. G. Baier**, et al. [200 Authors]
[Physical Review D](#), **110**, 4, (2024)
6. *Search for Gravitational-lensing Signatures in the Full Third Observing Run of the LIGO–Virgo Network.*
 R. Abbott, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal](#), **970**, 2, (2024)
5. *Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_{\odot}$ Compact Object and a Neutron Star.*
 A. G. Abac, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal Letters](#), **970**, 2, (2024)
4. *The NANOGrav 15 yr Data Set: Search for Transverse Polarization Modes in the Gravitational-wave Background.*

- Gabriella Agazie, [...], **J. G. Baier**, et al. [100 Authors]
[The Astrophysical Journal Letters](#), **964**, 1, (2024)
3. *Search for subsolar-mass black hole binaries in the second part of Advanced LIGO's and Advanced Virgo's third observing run.*
 LVK Collaboration, [...], **J. G. Baier**, et al. [200 Authors]
[Monthly Notices of the Royal Astronomical Society](#), **524**, 4, (2023)
 2. *Open Data from the Third Observing Run of LIGO, Virgo, KAGRA, and GEO.*
 R. Abbott, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal Supplements](#), **267**, 2, (2023)
 1. *Model-based Cross-correlation Search for Gravitational Waves from the Low-mass X-Ray Binary Scorpius X-1 in LIGO O3 Data.*
 R. Abbott, [...], **J. G. Baier**, et al. [200 Authors]
[The Astrophysical Journal Letters](#), **941**, 2, (2022)

Submitted.....

4. *Directional Anisotropic Sensitivity Curves for Pulsar Timing Arrays.*
 Daniel J. Oliver, Jeffrey S. Hazboun, Martine Maggi, **J. G. Baier**, Kyle E. Goullie
[Arxiv:2608.00250](#)
3. *No persuasive evidence yet of gravitational-wave tails from perturbations along the line of sight in LVK observations.*
 Kuba Kopczuk, [...], **J. G. Baier**, et al. [10 Authors]
[Arxiv:2607.12043](#)
2. *Mitigating the Timing Impact of Anomalous Pulse Profile Shape Variability in PSR J1713+0747 with Gaussian Component Modeling.*
 Shania A. Nichols, [...], **J. G. Baier**, et al. [34 Authors]
[Arxiv:2607.12038](#)
1. *The NANOGrav 15 yr Data Set: Impacts of Customized Chromatic Noise Models on Gravitational Wave Analyses.*
 Nikita Agarwal, [...], **J. G. Baier**, et al. [123 Authors]
[Arxiv:2606.28554](#)

In Prep.....

3. *A solar wind mean function for modeling delays in pulsar timing.*
 Jeremy Baier, Jeff Hazboun
[In Preparation](#)
2. *The NICER 7.9 year dataset – A Detection of Red Noise in B1937-21.*
 Kalista Wayt, **J. G. Baier**, Jeff Hazboun, The NICER Collaboration
[In Preparation](#)
1. *Improvements to Calculations of Stochastic GWB Sensitivity Curves.*
 Daniel and Oliver, Kyle Goullie, Martine Maggi, **J. G. Baier**, Jeff Hazboun
[In Preparation](#)